

CUSTOMER MILESTONE BRIEF

MS #24 - PL/I Discovery Conformance Lab

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.24.0
Date	2026-08-27
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #24 - PL/I Discovery Conformance Lab converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Replaces plausible text matching with a bounded, testable PL/I discovery capability and explicit unsupported cases.

Why this matters to a customer

Replaces plausible text matching with a bounded, testable PL/I discovery capability and explicit unsupported cases.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Replaced the PL/I language pack's line-oriented pattern extraction boundary with a tokenized, statement-aware supported-subset front end that preserves source locations.
- Published a content-addressed PL/I support matrix spanning 22 construct categories and explicit unsupported syntax.
- Added a synthetic 27-case conformance corpus with 25 program cases, two include cases, five expected blockers, and seven mutation-oriented cases.
- Added deterministic golden parse/reference results and a graph-bound coverage receipt recording recognized constructs, explicit gap codes, provenance, and claim boundaries.

- Made comments, strings, casing, spacing, continuation lines, missing includes, shadowed calls, malformed comments, unsupported preprocessors, and unsupported storage fail safely.
- Upgraded the PL/I pack contract to v1.2 and made the unified capability view depend on the conformance receipt before PL/I discovery can remain ready.
- Added PL/I breadth metrics to the customer/auditor projection while explicitly labelling the corpus synthetic, non-customer, and non-runtime evidence.
- Added Bash and PowerShell conformance entry points plus fail-closed tamper, graph-drift, and overclaim tests.
- Kept authorized z/OS execution, IBM compiler equivalence, arbitrary Enterprise PL/I support, mainframe equivalence, and production readiness blocked.

How it differs from earlier work

MS #24 builds on MS #23 (Stable Composite Estate and Developer Golden Path). The earlier milestone established its own bounded capability; this milestone adds pl/i discovery conformance lab while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use pl/i discovery conformance lab as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.24.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.